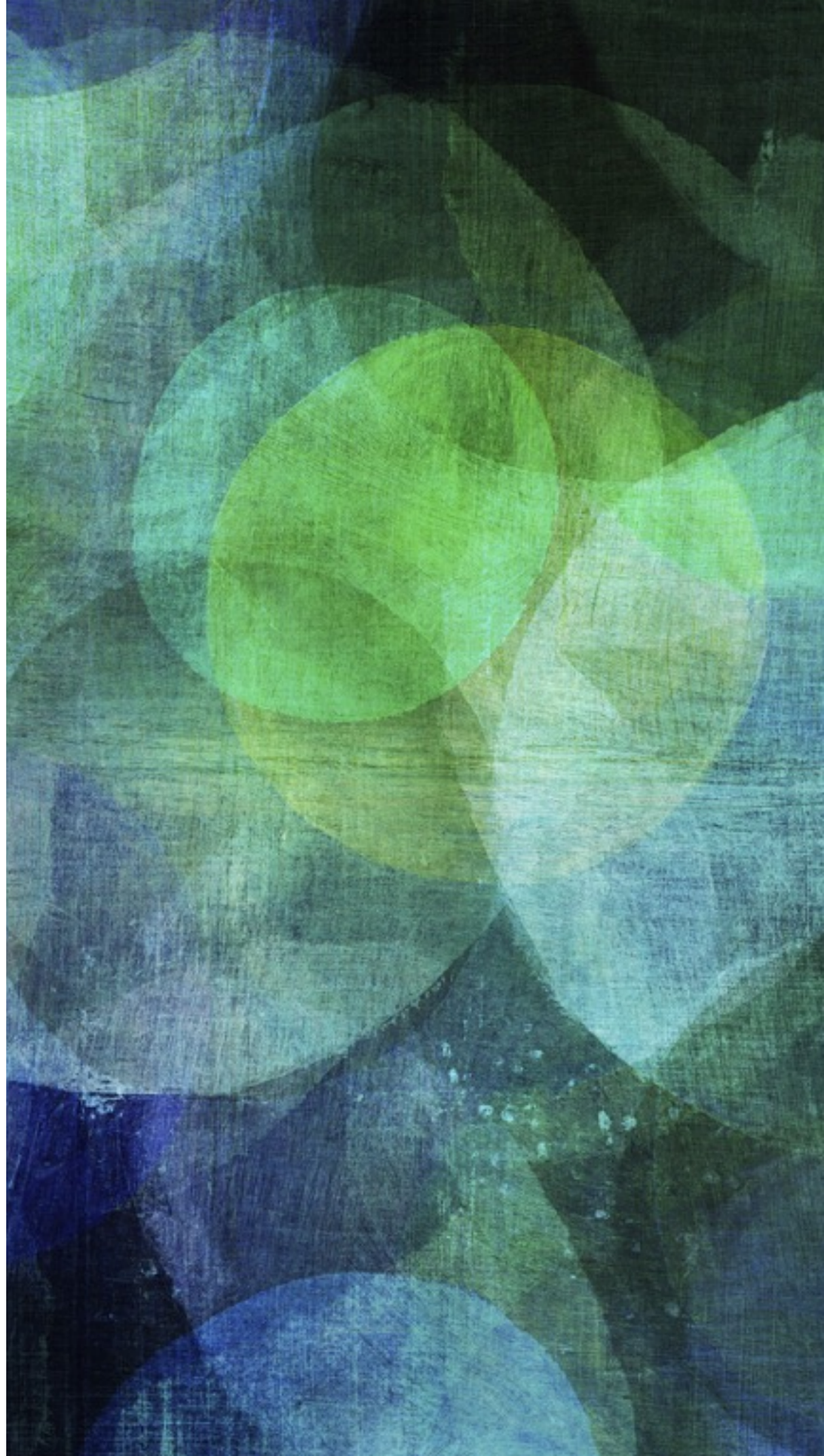
Carpentier, MV: **Revealing Graph Bandits for Maximising Local Influence**, AISTATS 2016

Wen, Kveton, MV: **Influence Maximization with Semi-Bandit Feedback,** (arXiv:1605.06593)

INFLUENCE MAXIMISATION

looking for the influential nodes
while exploring the graph



HOW TO RULE THE WORLD?

informatics mathematics

Influence the influential!

JULY 18, 2016

Religion

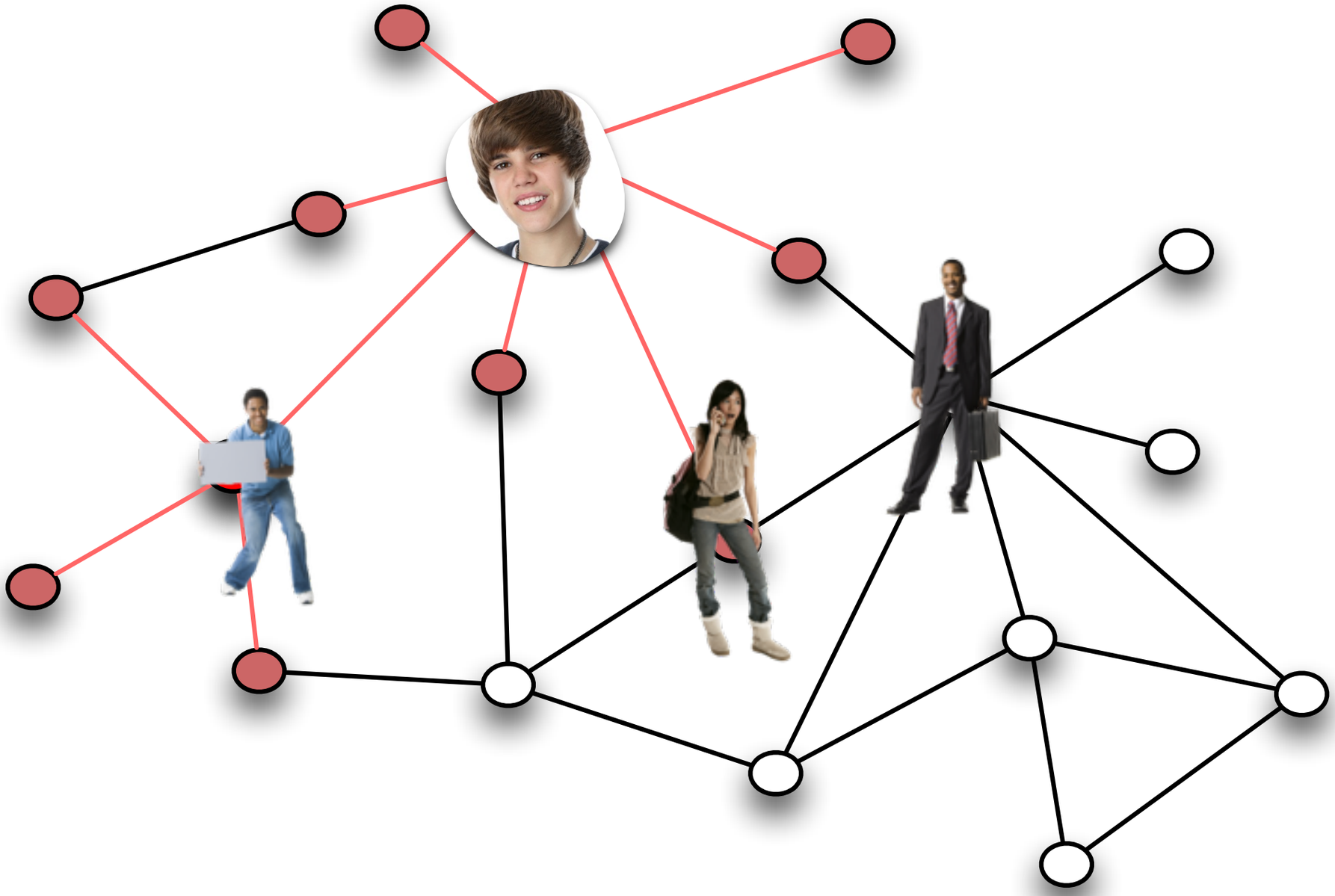
March 26, 2017

Politics

September 1, 2009

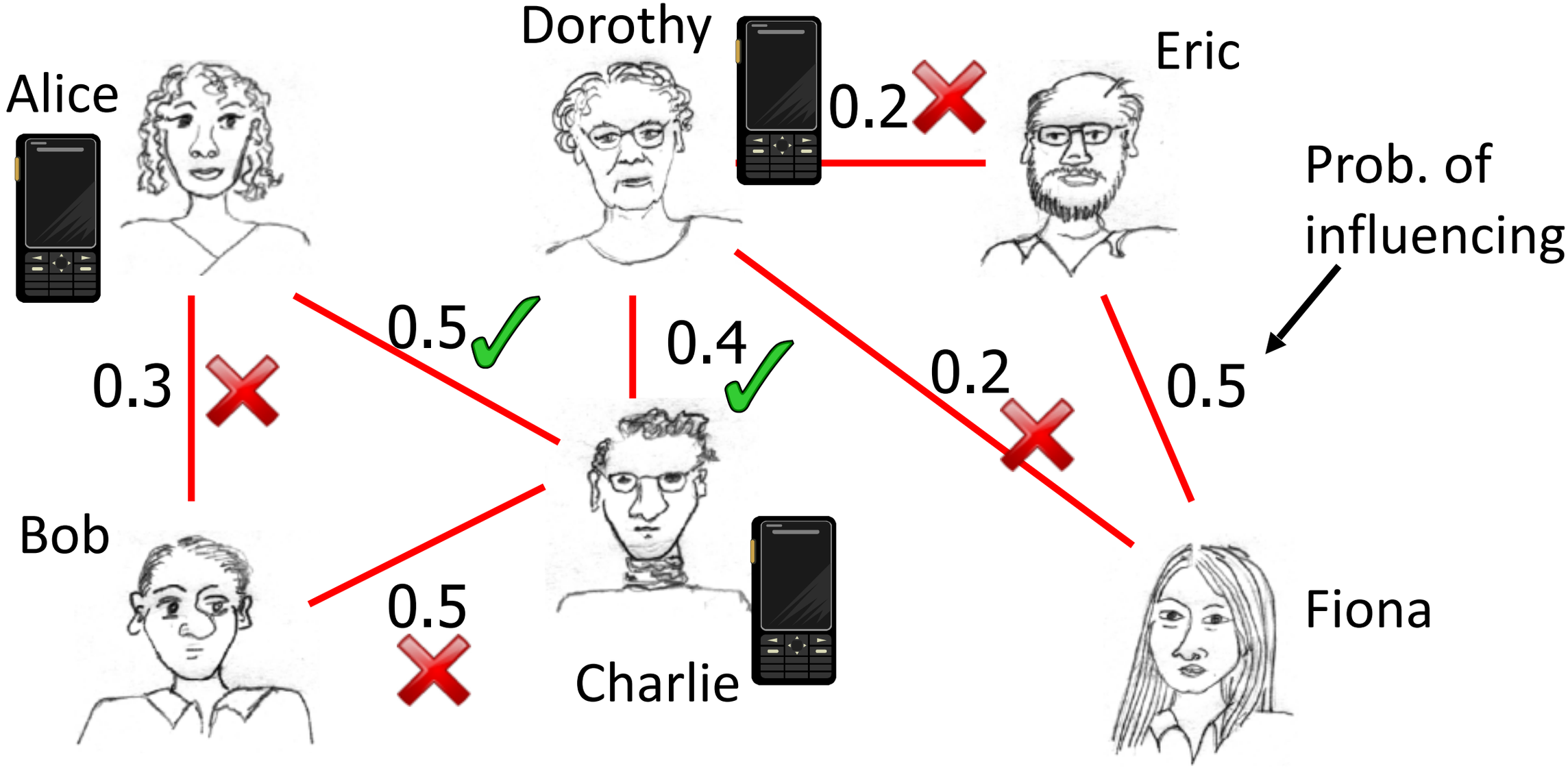
Culture

338 ET 200



$$F(S) = \text{spread}$$

EXAMPLE: INFLUENCE IN SOCIAL NETWORKS [KEMPE, KLEINBERG, TARDOS KDD '03]

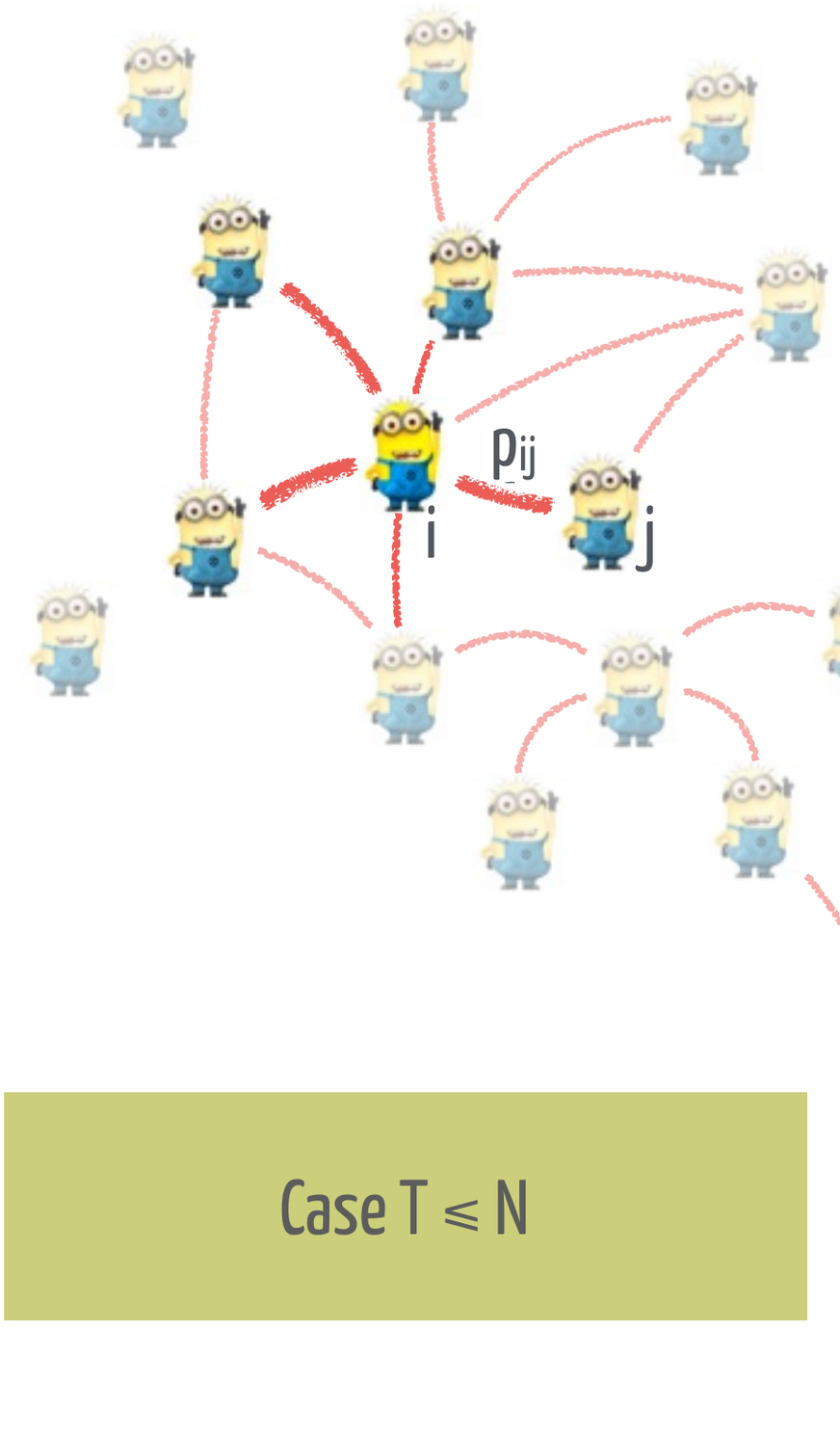


Who should get free cell phones?

V = {Alice,Bob,Charlie,Dorothy,Eric,Fiona}

F(A) = Expected number of people influenced when targeting A

REVEALING BANDITS FOR LOCAL INFLUENCE



Unknown $(p_{ij})_{ij}$ — (symmetric) probability of influences

In each time step t = 1, …., T

learner picks a node k_t

environment **reveals** the set of influenced node S_{kt}

Select influential people = Find the strategy maximising

$$L_T = \sum_{t=1}^{T} |S_{k_t,t}|$$

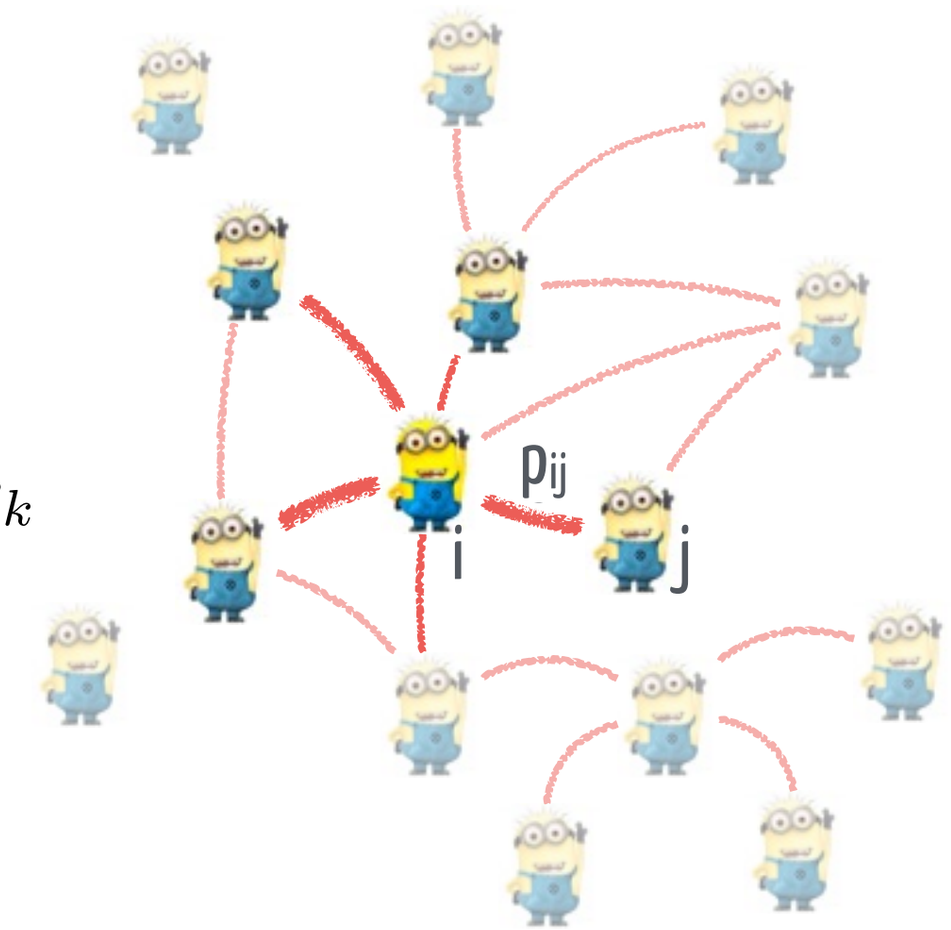
Why this is a **bandit problem**?

Case T ⩽ N

PERFORMANCE CRITERION

The number of expected influences of node **k** is by definition

$$r_k = \mathbb{E}\left[|S_{k,t}|\right] = \sum_{j \leq N} p_{k,j}$$



Oracle strategy always selects the best

$$k^\star = \arg\max_k \mathbb{E}\left[\sum_{t=1}^{T} |S_{k,t}|\right] = \arg\max_k Tr_k$$

Expected regret of the oracle strategy

$$\mathbb{E}\left[L_T^\star\right] = Tr_\star$$

Expected regret of any adaptive strategy unaware of $(p_{ij})_{ij}$

$$\mathbb{E}\left[R_T\right] = \mathbb{E}\left[L_T^\star\right] - \mathbb{E}\left[L_T\right]$$

BASELINE

- We **only** receive |S| instead of S
- Can be mapped to **multi-arm** bandits
  - rewards are 0, ..., N
  - variance bounded with r_{kt}

- We adapt **MOSS** to GraphMOSS
- Regret upper bound of GraphMOSS

each node at least once

$$\mathbb{E}\left[R_T\right] \leq U \min\left(r_\star T, r_\star N + \sqrt{r_\star T N}\right)$$

unlearnable case T ≤ N

- matching lower bound

Crash course on stochastic bandits?

GRAPHMOSS FOR THE RESTRICTED SETTING

GraphMOSS

Input

d: the number of nodes

n: time horizon

Initialization

Sample each arm twice

Update $\widehat{r}_{k,2d}$, $\widehat{\sigma}_{k,2d}$, and $T_{k,2d} \leftarrow 2$, for $\forall k \leq d$

for $t = 2d+1, \ldots, n$ **do**

$$C_{k,t} \leftarrow 2\widehat{\sigma}_{k,t}\sqrt{\frac{\max(\log(n/(dT_{k,t})),0)}{T_{k,t}}} + \frac{2\max(\log(n/(dT_{k,t})),0)}{T_{k,t}}, \text{ for } \forall k \leq d$$

$k_t \leftarrow \arg\max_k \widehat{r}_{k,t} + C_{k,t}$

Sample node k_t and receive $|S_{k_t,t}|$

Update $\widehat{r}_{k,t+1}$, $\widehat{\sigma}_{k,t+1}$, and $T_{k,t+1}$, for $\forall k \leq d$

end for

BACK TO THE REAL SETTING

- Can we actually do better?
  - Well, not really.....
  - Minimax optimal rate is still the same
- But the bad cases are somehow pathological
  - isolated nodes
  - uncorrelated being influenced and being influential
  - Barabási–Albert etc tell us that the real-world graphs are not like that
- Let's think of some measure of difficulty
  - to define some non-degenerate cases
  - ideas?

DETECTABLE DIMENSION

- number of nodes we can efficiently extract in less than n rounds
- function D controls number of nodes given a gap

$$D(\Delta) = |\{i \leq N : r_\star^\circ - r_i^\circ \leq \Delta\}|$$

- D(r) = N for r≥ r* and D(0) = number of most influenced nodes
- **Detectable dimension D*** = D(Δ*)
- **Detectable gap Δ*** constants coming from the analysis and the Bernstein inequality

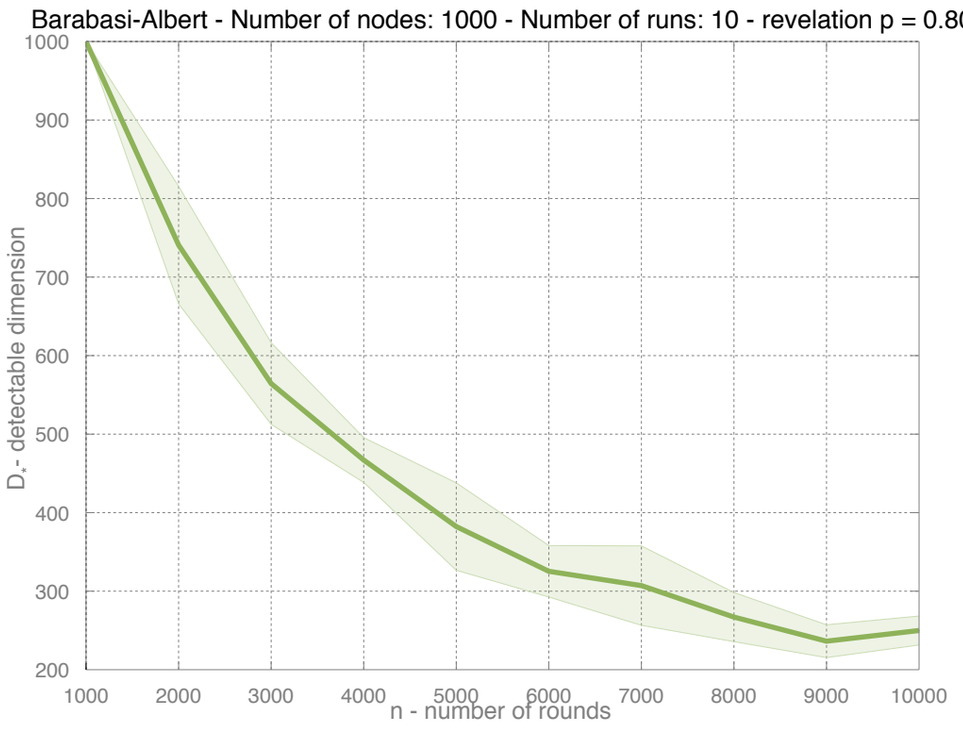
$$\Delta_\star = 16\sqrt{\frac{r_\star^\circ N \log(TN)}{T_\star}} + \frac{144N\log(TN)}{T_\star}$$

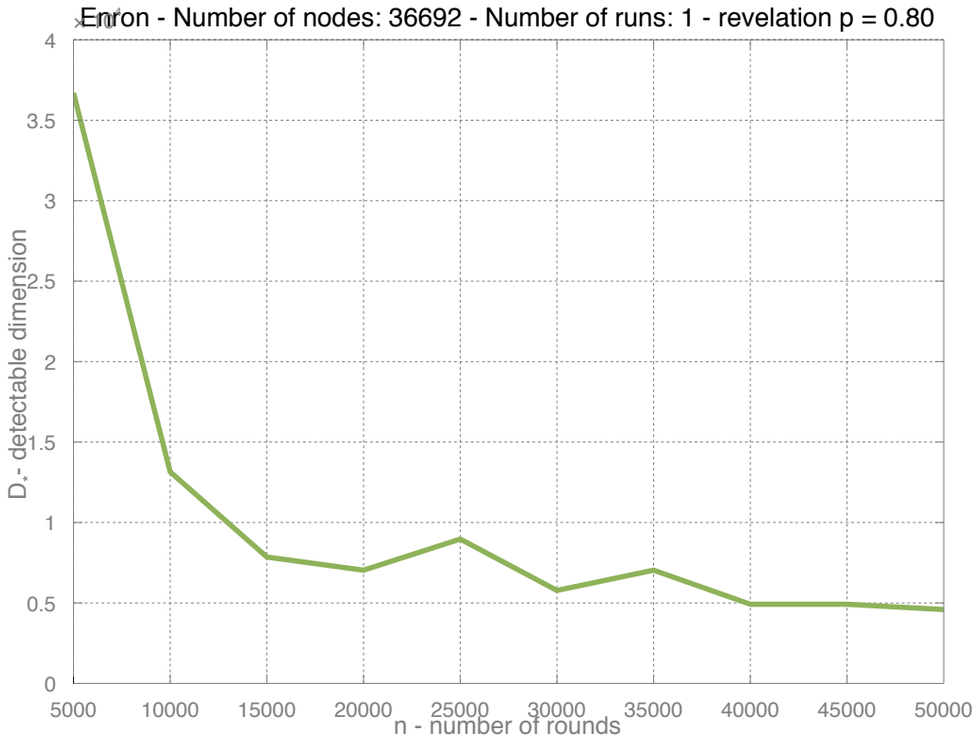
- **Detectable horizon T*, smallest integer s.t.** $T_\star r_\star^\circ \geq \sqrt{D_\star T r_\star^\circ}$,
- **Equivalently: D* corresponding to smallest T* such that**

$$T_\star r_\star^\circ \geq \sqrt{D\left(16\sqrt{\frac{r_\star^\circ N\log(TN)}{T_\star}} + \frac{144N\log(TN)}{T_\star}\right) T r_\star^\circ}$$

HOW DOES D* BEHAVE?

- For (easy, structured) star graphs $D_* = 1$ even for small n (big gain)
- For (difficult) empty graphs $D_* = N$ even for large T (no gain)
- In general: D_* roughly decreases with n and it is small when D decreases quickly
- For n large enough D_* is the number of the most influences nodes
- Example: D_* for Barabási–Albert model & Enron graph as a function of T





BARE SOLUTION

BAndit REvelator: 2-phase algorithm

- **global** exploration phase
  - super-efficient exploration 😺
  - linear regret 😿 — needs to be short!
  - extracts D⋆ nodes
- **bandit** phase
  - uses a minimax-optimal bandit algorithm
  - GraphMOSS is a little brother of MOSS
  - has a "square root" regret on D⋆ nodes
- **D⋆ realizes the optimal trade-off !**
  - different from exploration/exploitation tradeoff

Algorithm

BARE - BAndit REvelator

Input

d: the number of nodes

n: time horizon

Initialization

$T_{k,t} \leftarrow 0$, for $\forall k \leq d$

$\widehat{r^{\circ}_{k,t}} \leftarrow 0$, for $\forall k \leq d$

$t \leftarrow 1$, $\widehat{T}_\star \leftarrow 0$, $\widehat{D}_{\star,t} \leftarrow d$, $\widehat{\sigma}_{\star,1} \leftarrow d$

Global exploration phase

while $t\left(\widehat{\sigma}_{\star,t} - 4\sqrt{d\log(dn)/t}\right) \leq \sqrt{\widehat{D}_{\star,t} n}$ **do**

Influence a node at random (choose k_t uniformly at random) and get $S_{k_t,t}$ from this node

$\widehat{r^{\circ}_{k,t+1}} \leftarrow \frac{t}{t+1}\widehat{r^{\circ}_{k,t}} + \frac{d}{t+1} S_{k_t,t}(k)$

$\widehat{\sigma}_{\star,t+1} \leftarrow \max_{k'} \sqrt{\widehat{r^{\circ}_{k',t+1}} + 8d\log(nd)/(t+1)}$

$w_{\star,t+1} \leftarrow 8\widehat{\sigma}_{\star,t+1}\sqrt{\frac{d\log(nd)}{t+1}} + \frac{24d\log(nd)}{t+1}$

$\widehat{D}_{\star,t+1} \leftarrow \left|\left\{k : \max_{k'} \widehat{r^{\circ}_{k',t+1}} - \widehat{r^{\circ}_{k,t+1}} \leq w_{\star,t+1}\right\}\right|$

$t \leftarrow t+1$

end while

$\widehat{T}_\star \leftarrow t$.

Bandit phase

Run minimax-optimal bandit algorithm on the $\widehat{D}_{\star,\widehat{T}_\star}$ chosen nodes (e.g., Algorithm 1)

EMPIRICAL RESULTS

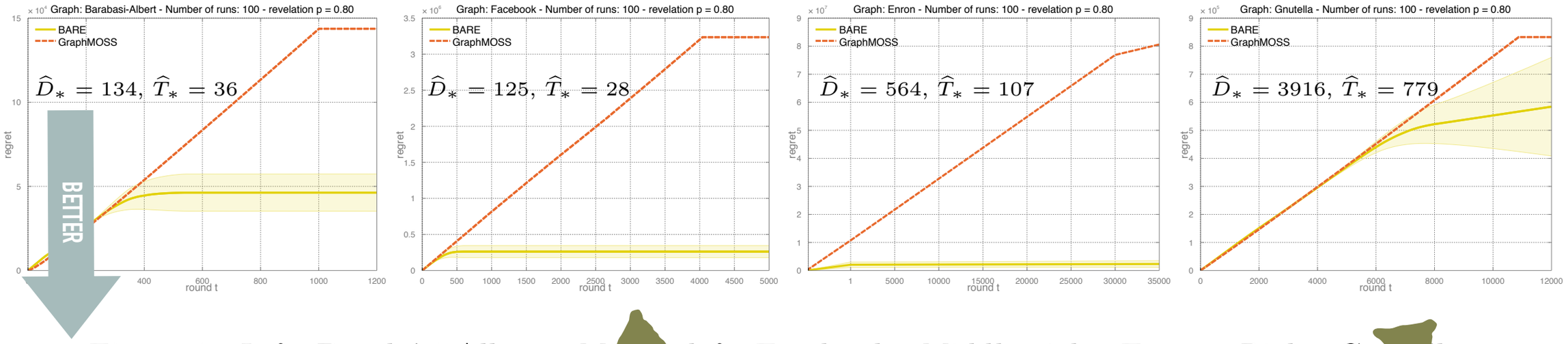
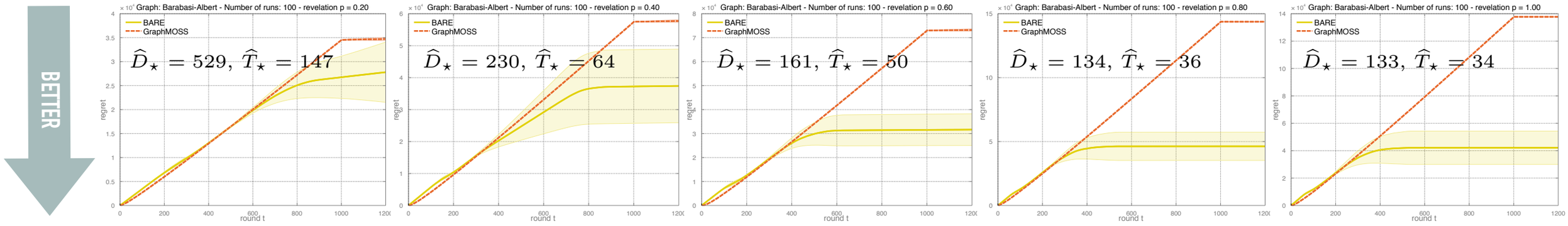


Figure 1: *Left*: Barabási-Albert. *Middle left*: Facebook. *Middle right*: Enron. *Right*: Gnutella.

Enron and Facebook **vs.** Gnutella (decentralised)



Varying a (constant) probability of influence

REVEALING BANDITS: WHAT DO YOU MEAN?

- Ignoring the structure again?
- **BAndit REvelator:** 2-phase algorithm
- **global** exploration phase
  - super-efficient exploration
  - linear regret — needs to be short!
  - extracts **D*** nodes
- **bandit** phase
  - uses a minimax-optimal bandit algorithm (GraphMOSS)
  - has a "square root" regret on **D*** nodes
- **D*** realizes the optimal trade-off !
  - different from exploration/exploitation tradeoff

$$\widetilde{\mathcal{O}}\left(\sqrt{r_* T N}\right)$$

reward of the best node

Regret of BARE

$$\mathcal{O}(\sqrt{r_\star T D_\star})$$

- **D*** - detectable dimension (depends on T and the structure)
  - **good case**: star-shaped graph can have D* = 1
  - **bad case:** a graph with many small cliques.
  - **the worst case:** all nodes are disconnected except 2

NEXT: GLOBAL INFLUENCE MODELS

- Kempe, Kleinberg, Tárdos, 2003, 2015: **Independence Cascades**, Linear Threshold models
  - **global** and **multiple-source** models
- Different feed-back models
  - **Full bandit** (only the number of influenced nodes)
  - **Node-level semi-bandit** (identities of influenced nodes)
  - **Edge-level semi-bandit** (identities of influenced edges)
    - Wen, Kveton, Valko, Vaswani, **NIPS 2017**
    - IMLinUCB with linear parametrization of edge weights
    - Regret analysis for **general graphs, cascading model, and multiple-sources**

Online Influence Maximization under Independent Cascade Model with Semi-Bandit Feedback

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Presented 1 year ago
at NIPS 2017, Long Beach, CA

Abstract

We study the stochastic online problem of learning to influence in a social network with semi-bandit feedback, where we observe how users influence each other. The problem combines challenges of limited feedback, because the learning agent only observes the influenced portion of the network, and combinatorial number of actions, because the cardinality of the feasible set is exponential in the maximum number of influencers. We propose a computationally efficient UCB-like algorithm, IMLinUCB, and analyze it. Our regret bounds are polynomial in all quantities of interest; *reflect the structure of the network* and the *probabilities of influence*. Moreover, they do not depend on inherently large quantities, such as the cardinality of the action set. To the best of our knowledge, these are the first such results. IMLinUCB permits linear generalization and therefore is suitable for large-scale problems. Our experiments show that the regret of IMLinUCB scales as suggested by our upper bounds in several representative graph topologies; and based on linear generalization, IMLinUCB can significantly reduce regret of real-world influence maximization semi-bandits.

CHALLENGES AND SOLUTIONS

- **Already the offline problem is NP hard**
  - solution: **approximation/randomized algorithms**
- **Lots of edges**
  - lots of parameters to learn, if we want to scale, we need to reduce this complexity
  - solution: **linear approximation of probabilities**
- **Combinatorial size of possible seed-sets**
  - Combinatorial Bandits: IMLinUCB
- **Understanding what's going on?**
  - known analyses VERY loose (e.g., scaling with $1/p_{min}$, or only assymptotic)

seed set

seed size

$$\max_{\mathcal{S}:\,|\mathcal{S}|=K} f(\mathcal{S}, \overline{w})$$

APPROXIMATION ORACLE

- the optimal offline solution

seed size

$$\max_{\mathcal{S}:\,|\mathcal{S}|=K} f(\mathcal{S}, \overline{w})$$

- the oracle solution that is γ-optimal with probability at least α

$$\mathcal{S}^* = \texttt{ORACLE}(\mathcal{G}, K, \overline{w})$$

- γ-optimal

$$f(\mathcal{S}^*, \overline{w}) \;\geq\; \gamma f(\mathcal{S}^{\text{opt}}, \overline{w})$$

- γ-optimal with probability at least α

$$\mathbb{E}\left[f(\mathcal{S}^*, \overline{w})\right] \geq \alpha\gamma f(\mathcal{S}^{\text{opt}}, \overline{w})$$

- Our problem is a triple:

$$(\mathcal{G}, K, \overline{w})$$

unknown to the agent

topology

seed size

LINEAR GENERALIZATION

— learning the only network (weights) is VERY impractical

$$\rho \stackrel{\Delta}{=} \max_{e \in \mathcal{E}} |\overline{w}(e) - x_e^{\mathsf{T}} \theta^*|$$

this is small

true weights

linear approximation

— by choosing the dimension (size of θ^*) we can reduce this complexity

— if we do not want to lose generality we set **d** to the number of edges

ALGORITHM AND PERFORMANCE MEASURE

Algorithm 1 `IMLinUCB`: Influence Maximization Linear UCB

Input: graph $\mathcal{G}$, source node set cardinality K, oracle `ORACLE`, feature vector x_e's, and algorithm parameters $\sigma, c > 0$,

Initialization: $B_0 \leftarrow 0 \in \Re^d$, $\mathbf{M}_0 \leftarrow I \in \Re^{d \times d}$

for $t = 1, 2, \ldots, n$ **do**

1. set $\overline{\theta}_{t-1} \leftarrow \sigma^{-2}\mathbf{M}_{t-1}^{-1}B_{t-1}$ and the UCBs as $U_t(e) \leftarrow \mathrm{Proj}_{[0,1]}\left(x_e^\intercal \overline{\theta}_{t-1} + c\sqrt{x_e^\intercal \mathbf{M}_{t-1}^{-1} x_e}\right)$ for all $e \in \mathcal{E}$
2. choose $\mathcal{S}_t \in \texttt{ORACLE}(\mathcal{G}, K, U_t)$, and observe the edge-level semi-bandit feedback
3. update statistics:
   (a) initialize $\mathbf{M}_t \leftarrow \mathbf{M}_{t-1}$ and $B_t \leftarrow B_{t-1}$
   (b) for all observed edges $e \in \mathcal{E}$, update $\mathbf{M}_t \leftarrow \mathbf{M}_t + \sigma^{-2}x_e x_e^\intercal$ and $B_t \leftarrow B_t + x_e \mathbf{w}_t(e)$

$$R^\eta(n) = \sum_{t=1}^{n} \mathbb{E}\left[R_t^\eta\right]$$

scaled regret

$$R_t^\eta = f(\mathcal{S}^{\mathrm{opt}}, \mathbf{w}_t) - \frac{1}{\eta} f(\mathcal{S}_t, \mathbf{w}_t)$$

MAXIMUM OBSERVED RELEVANCE

$N_{\mathcal{S},e} \triangleq \sum_{v \in \mathcal{V} \setminus \mathcal{S}} \mathbf{1}\{e \text{ is relevant to } v \text{ under } \mathcal{S}\}$ and $P_{\mathcal{S},e} \triangleq \mathbb{P}(e \text{ is observed} \mid \mathcal{S})$

only depends on topology

depends on both

$$C_* \triangleq \max_{\mathcal{S}:\,|\mathcal{S}|=K} \sqrt{\sum_{e \in \mathcal{E}} N_{\mathcal{S},e}^2 P_{\mathcal{S},e}}$$

max (over) 2-norm of N weighted by P

- Worst-case upper bound:

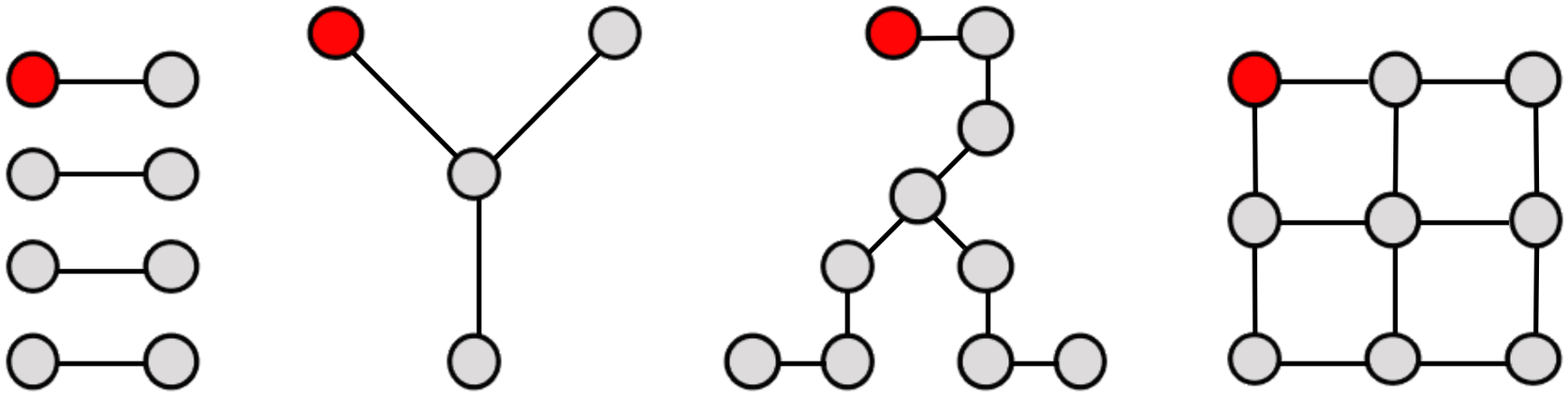
#nodes

#edges

$$C_* \leq C_{\mathcal{G}} \triangleq \max_{\mathcal{S}:\,|\mathcal{S}|=K} \sqrt{\sum_{e \in \mathcal{E}} N_{\mathcal{S},e}^2} \leq (L-K)\sqrt{|\mathcal{E}|} = \mathcal{O}\left(L\sqrt{|\mathcal{E}|}\right) = \mathcal{O}\left(L^2\right)$$

seed size

WORST-CASE BOUNDS



topology	$C_{\mathcal{G}}$ (worst-case C_*)	$R^{\alpha\gamma}(n)$ for general $\mathbf{X}$	$R^{\alpha\gamma}(n)$ for $\mathbf{X} = \mathbf{I}$
bar graph	$\mathcal{O}(\sqrt{K})$	$\widetilde{\mathcal{O}}\left(dK\sqrt{n}/(\alpha\gamma)\right)$	$\widetilde{\mathcal{O}}\left(L\sqrt{Kn}/(\alpha\gamma)\right)$
star graph	$\mathcal{O}(L\sqrt{K})$	$\widetilde{\mathcal{O}}\left(dL^{\frac{3}{2}}\sqrt{Kn}/(\alpha\gamma)\right)$	$\widetilde{\mathcal{O}}\left(L^2\sqrt{Kn}/(\alpha\gamma)\right)$
ray graph	$\mathcal{O}(L^{\frac{5}{4}}\sqrt{K})$	$\widetilde{\mathcal{O}}\left(dL^{\frac{7}{4}}\sqrt{Kn}/(\alpha\gamma)\right)$	$\widetilde{\mathcal{O}}\left(L^{\frac{9}{4}}\sqrt{Kn}/(\alpha\gamma)\right)$
tree graph	$\mathcal{O}(L^{\frac{3}{2}})$	$\widetilde{\mathcal{O}}\left(dL^2\sqrt{n}/(\alpha\gamma)\right)$	$\widetilde{\mathcal{O}}\left(L^{\frac{5}{2}}\sqrt{n}/(\alpha\gamma)\right)$
grid graph	$\mathcal{O}(L^{\frac{3}{2}})$	$\widetilde{\mathcal{O}}\left(dL^2\sqrt{n}/(\alpha\gamma)\right)$	$\widetilde{\mathcal{O}}\left(L^{\frac{5}{2}}\sqrt{n}/(\alpha\gamma)\right)$
complete graph	$\mathcal{O}(L^2)$	$\widetilde{\mathcal{O}}\left(dL^3\sqrt{n}/(\alpha\gamma)\right)$	$\widetilde{\mathcal{O}}\left(L^4\sqrt{n}/(\alpha\gamma)\right)$

Table 1: $C_{\mathcal{G}}$ and *worst-case* regret bounds for different graph topologies

RESULTS

$$R^{\alpha\gamma}(n) \leq \frac{2cC_*}{\alpha\gamma}\sqrt{dn|\mathcal{E}|\log_2\left(1+\frac{n|\mathcal{E}|}{d}\right)}+1 = \widetilde{\mathcal{O}}\left(dC_*\sqrt{|\mathcal{E}|n}/(\alpha\gamma)\right)$$
$$\leq \widetilde{\mathcal{O}}\left(d(L-K)|\mathcal{E}|\sqrt{n}/(\alpha\gamma)\right).$$

How good (tight) is this?

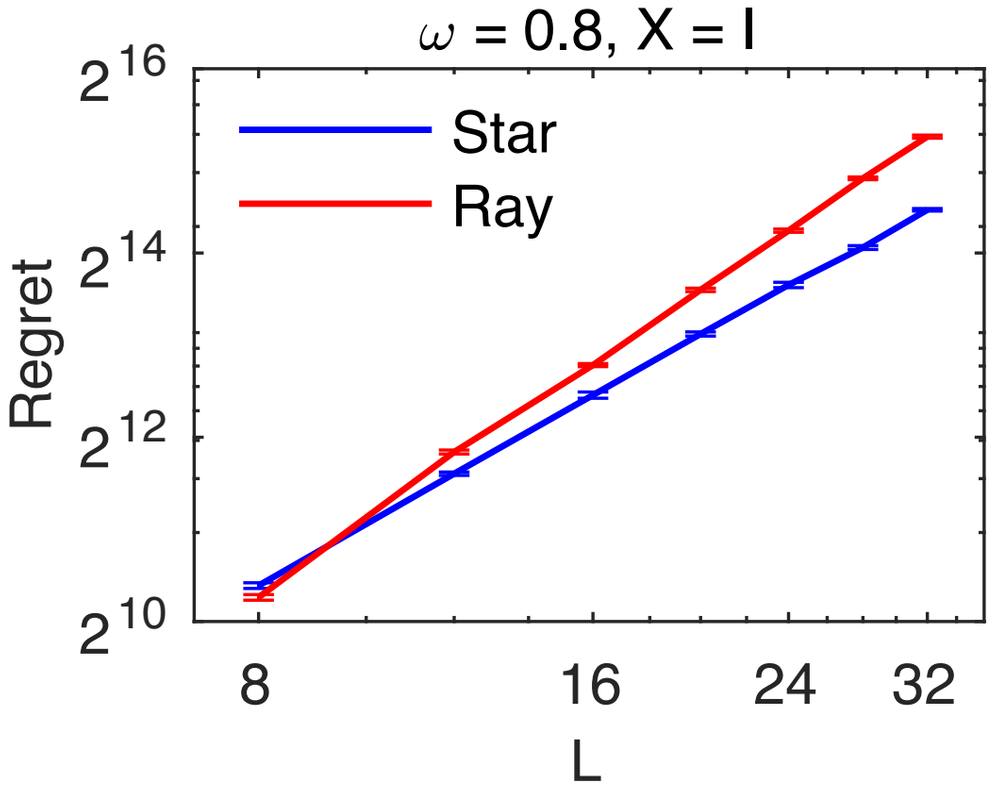
- comparison with linear bandits
- comparison with general combinatorial bandits
- (L-K) factor
- **How good is C*?**

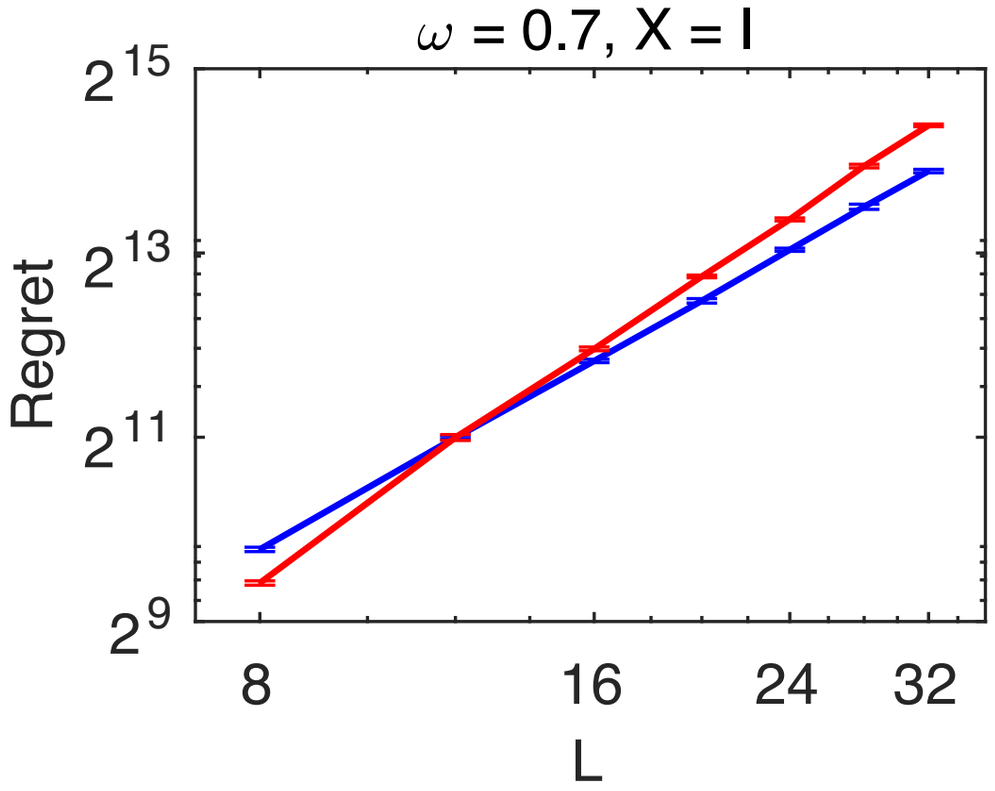
NUMERICAL EVALUATION

- **Objective:** "Check" how good is our C*
- Tabular case, K = 1, exact comparison possible, all weights are same = ω

Star $\quad \widetilde{\mathcal{O}}(L^2)$ vs. $\mathcal{O}(L^{2.040})$ and $\mathcal{O}(L^{2.056})$

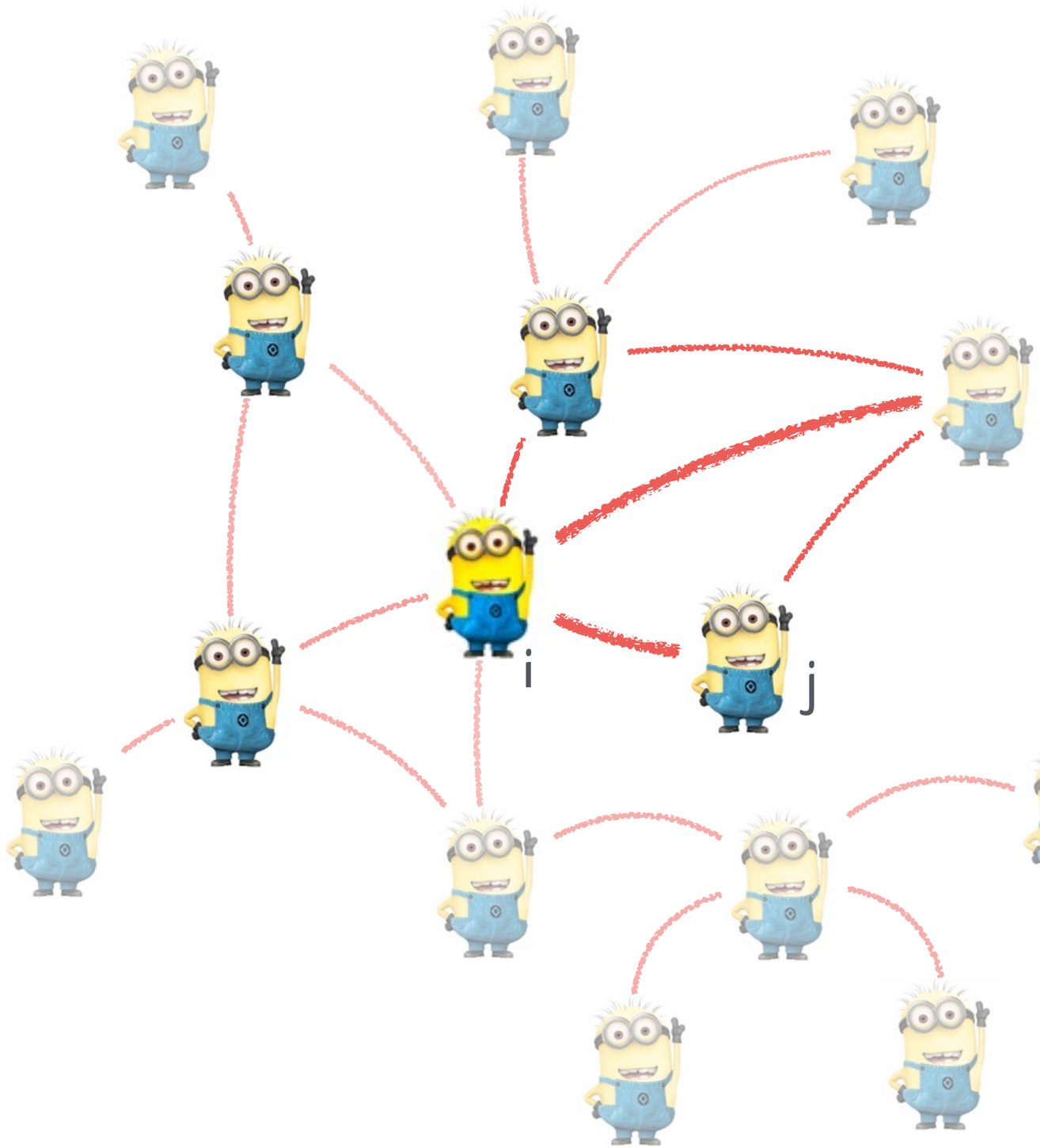
Ray $\quad \widetilde{\mathcal{O}}(L^{\frac{9}{4}})$ vs. $\mathcal{O}(L^{2.488})$ and $\mathcal{O}(L^{2.467})$





- **Conclusion:** evidence that our C* is a reasonable complexity measure

WHERE IS THE CHALLENGE FOR THE ANALYSIS?



How much we are losing using **UCBs** instead of the **true influence function**?

PROOF SKETCH

- when are our upper bounds on the estimates right?

$$\xi_{t-1} = \{|x_e^\intercal(\overline{\theta}_{\tau-1} - \theta^*)| \leq c\sqrt{x_e^\intercal \mathbf{M}_{\tau-1}^{-1} x_e}, \forall e \in \mathcal{E}, \forall \tau \leq t\}$$

- .... decomposes the regret at round t

$$\mathbb{E}[R_t^{\alpha\gamma}] \leq \mathbb{P}(\xi_{t-1})\,\mathbb{E}[R_t^{\alpha\gamma}|\xi_{t-1}] + \mathbb{P}(\overline{\xi}_{t-1})\,[L-K]$$

- monotonicity of f

$$\mathbb{E}[R_t^{\alpha\gamma}|\xi_{t-1}] \leq \mathbb{E}[f(\mathcal{S}_t, U_t) - f(\mathcal{S}_t, \overline{w})|\xi_{t-1}]\,/(\alpha\gamma)$$

- linearity of expectation

decomposed into nodes

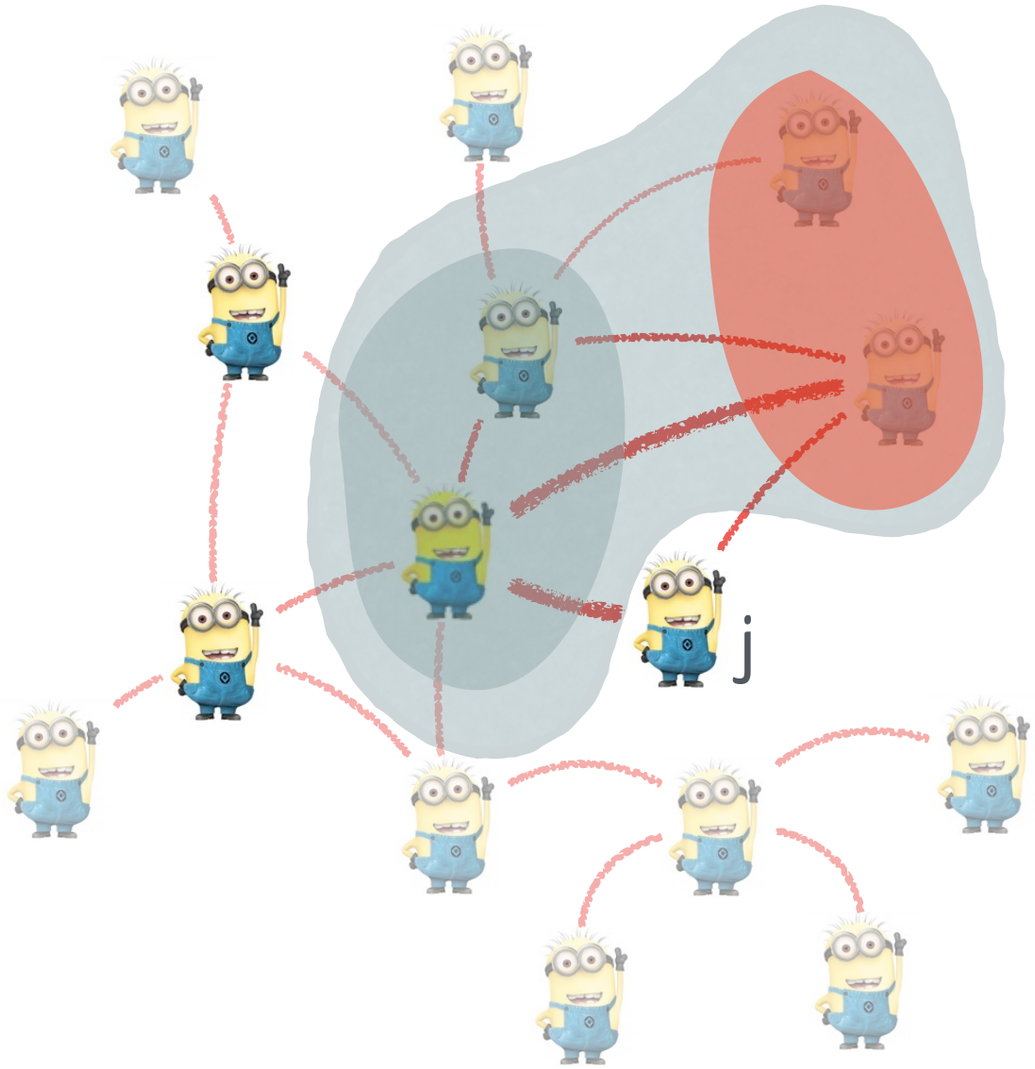
$$f(\mathcal{S}_t, U_t) - f(\mathcal{S}_t, \overline{w}) = \sum_{v \in \mathcal{V}\setminus\mathcal{S}_t} [f(\mathcal{S}_t, U_t, v) - f(\mathcal{S}_t, \overline{w}, v)]$$

- **difference of two Markov chains, is controlled by the edge-level gap**

$$f(\mathcal{S}_t, U_t, v) - f(\mathcal{S}_t, \overline{w}, v) \leq \sum_{e \in \mathcal{E}_{\mathcal{S}_t, v}} \mathbb{E}[\mathbf{1}\{O_t(e)\}\,[U_t(e) - \overline{w}(e)]|\mathcal{H}_{t-1}, \mathcal{S}_t]$$

probability that node v is influenced

DIFFUSION PROCESS OF A MARKOV CHAIN



- Sets of progressive diffusion
  - modeling diffusion steps
- Random stopping time
  - but bounded
- Topological ordering

$$\mathcal{S}^0 \triangleq \mathcal{S}_t$$

$$\mathcal{S}^{\tau+1} \triangleq \left\{ u_2 \in \mathcal{V}_{\mathcal{S}_t,v} : u_2 \notin \cup_{\tau'=0}^{\tau} \mathcal{S}^{\tau'} \text{ and } \exists e = (u_1, u_2) \in \mathcal{E}_{\mathcal{S}_t,v} \text{ s.t. } u_1 \in \mathcal{S}^{\tau} \text{ and } \mathbf{w}(e) = 1 \right\}$$

DIFFUSION PROCESS OF A MARKOV CHAIN

- additional influenced nodes in one diffusion step

$$h\left(\mathcal{S}^{\tau},\mathcal{S}^{0:\tau-1},U\right)-h\left(\mathcal{S}^{\tau},\mathcal{S}^{0:\tau-1},w\right)\leq\sum_{e\in\mathcal{E}(\mathcal{S}^{\tau},\mathcal{S}^{0:\tau})}[U(e)-w(e)]$$
$$+\mathbb{E}\left[h\left(\mathcal{S}^{\tau+1},\mathcal{S}^{0:\tau},U\right)-h\left(\mathcal{S}^{\tau+1},\mathcal{S}^{0:\tau},w\right)\middle|(\mathcal{S}^{\tau},\mathcal{S}^{0:\tau-1})\right]$$

- …. over a topological ordering

$$f(\mathcal{S}_t,U,v)-f(\mathcal{S}_t,w,v)\leq\mathbb{E}\left[\sum_{\tau=0}^{\widetilde{\tau}-1}\sum_{e\in\mathcal{E}(\mathcal{S}^{\tau},\mathcal{S}^{0:\tau})}[U(e)-w(e)]\middle|\mathcal{S}_t\right]$$

- **difference of two Markov chains, is controlled by the edge-level gap**

$$f(\mathcal{S}_t,U_t,v)-f(\mathcal{S}_t,\overline{w},v)\leq\sum_{e\in\mathcal{E}_{\mathcal{S}_t,v}}\mathbb{E}\left[\mathbf{1}\left\{O_t(e)\right\}[U_t(e)-\overline{w}(e)]|\mathcal{H}_{t-1},\mathcal{S}_t\right]$$

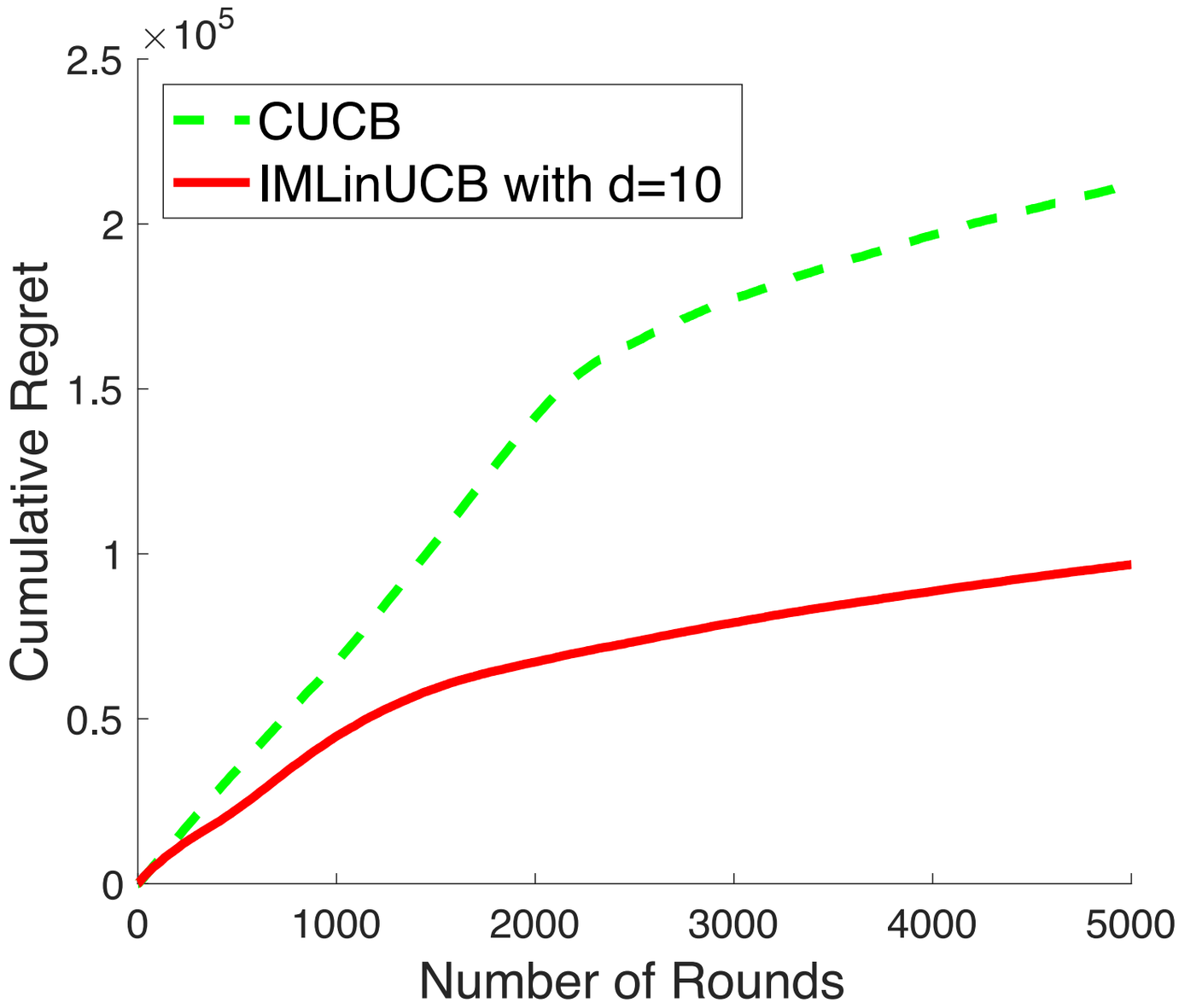
- **and we get the final result**

$$R^{\alpha\gamma}(n)\leq\frac{2c}{\alpha\gamma}\mathbb{E}\left[\sum_{t=1}^{n}\sum_{e\in\mathcal{E}}\mathbf{1}\{O_t(e)\}N_{\mathcal{S}_t,e}\sqrt{x_e^{\intercal}\mathbf{M}_{t-1}^{-1}x_e}\right]+[L-K]\sum_{t=1}^{n}\mathbb{P}\left(\overline{\xi}_{t-1}\right)$$

- …. and C* appears because

$$\sum_{t=1}^{n}\sum_{e\in\mathcal{E}}\mathbf{1}\{O_t(e)\}N_{\mathcal{S}_t,e}\sqrt{x_e^{\intercal}\mathbf{M}_{t-1}^{-1}x_e}\leq\sqrt{\left(\sum_{t=1}^{n}\sum_{e\in\mathcal{E}}\mathbf{1}\{O_t(e)\}N_{\mathcal{S}_t,e}^2\right)\frac{dE_*\log\left(1+\frac{nE_*}{d\sigma^2}\right)}{\log\left(1+\frac{1}{\sigma^2}\right)}}.$$

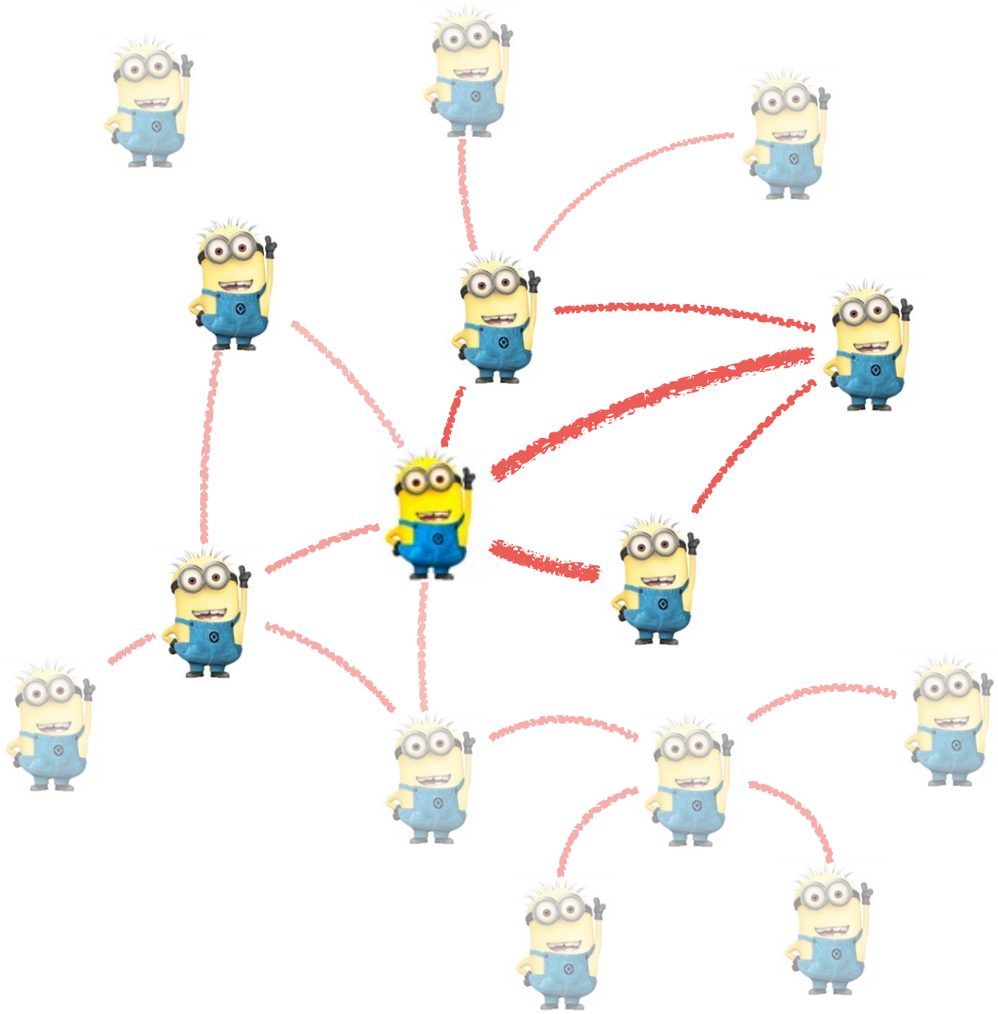
FACEBOOK EXPERIMENT



- real Facebook (a small subgraph)
- weights from U(0,0.1)
- **nodetovec** with d=10
  - imperfect
- K = 10
- CUCB with no linear generalisation

CONCLUSION AND NEXT STEPS

- **Active learning on graphs: online influence maximization**
  - learning the graph **while** acting on it optimally
  - global cascading model with edge level feedback
  - **difficulty of the problem** and scaling with it



- **What is next?**
  - node-level feedback
  - dynamic/evolving graphs
  - realistic accessibility constraints